



GOSFORD HIGH SCHOOL

MATHEMATICS – EXTENSION 1

2005

YEAR 11 PRELIMINARY YEARLY EXAM

Time allowed 2 hours (*Plus 5 minutes reading time*)

NAME :

TEACHER :

DIRECTIONS TO CANDIDATES

- **Attach this cover sheet** to your answers
- **All** questions may be attempted
- All necessary working should be shown in every question. Marks may not be awarded for careless or badly arranged work.
- **START** each question on a new page, clearly marked with your name and the question number
- Approved calculators and templates may be used
- Write using blue or black pen

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Question 1

- a) A is the point $(-5, 7)$ and B is $(-1, -2)$. Find the point P which divides AB externally in the ratio 3:1. 3
- b) Solve $x - 5 \geq \frac{6}{x}$ 3
- c) (i) Write down 3 formulae for $\cos 2\theta$ 1
- (ii) Prove that $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ 4

Question 2 Begin a new sheet of paper

- a) For what values of p will $x^2 - (p + 3)x + 4p = 0$ have:
- (i) Equal roots? 2
- (ii) No real roots? 2
- b) (i) Find the equation of the tangent to the parabola $y = 3x - x^2$ at the point where $x = 2$. 3
- (ii) Draw a sketch of the parabola and this tangent. 2
- c) Use the expansion of $\tan(A + B)$ to find the value of $\tan 75^\circ$ in simplest exact form. 2

Question 3 Begin a new sheet of paper

- a) Solve $3^{2x} + 2(3^x) - 15 = 0$ 4
- b) Find the values of A, B, and C if $7x^2 - 5x + 3 \equiv A(x + 1)^2 + B(x + 1) + C$ 3
- c) A parabola has its focus at $(-2, 3)$ and its directrix has equation $x = 4$.
- (i) Draw a sketch, and use it to find 1
- (ii) the coordinates of its vertex. 1
- (iii) the equation of the parabola. 3

Question 4 Begin a new sheet of paper**Marks**

a) Differentiate the following (Simplify your answer):

(i) $\frac{x}{\sqrt{1-2x}}$ 3

(ii) $\frac{2x^2 - 5x + 1}{\sqrt{x}}$ 3

b) Differentiate from first principles $f(x) = x - 5x^2$ 4c) Find the point on the curve $y = x^2 - 3x - 7$ where the tangent is parallel to the line $y = 5x - 4$. 2**Question 5** Begin a new sheet of papera) For the function $y = x^4 - 4x^3$ (i) Find any stationary points and determine their nature 4(ii) Find any points of inflexion 2(iii) Find the intercepts with the axes 1(iv) Sketch the curve 3(v) For what values of x is the curve concave down? 1b) Prove that the locus of points equidistant from $A(-5,4)$ and $B(3, 2)$ is the perpendicular bisector of AB . 4**Question 6** Begin a new sheet of papera) Solve $\sin 3x \cos x - \cos 3x \sin x = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$ 3b) (i) Express $2\sqrt{3} \cos x - 2 \sin x$ in the form $R \cos(x + \alpha)$ where R is positive and α is acute. 3(ii) Hence find the range of the function $f(x) = 2\sqrt{3} \cos x - 2 \sin x$ 1c) (i) If $t = \tan \frac{x}{2}$, show that $3 \sin x - 4 \cos x - 4 = \frac{6t - 8}{1 + t^2}$ 2(ii) Hence solve $3 \sin x - 4 \cos x - 4 = 0$ for $0^\circ \leq x \leq 360^\circ$ 3

Question 7 Begin a new sheet of paper**Marks**

- a) The quadratic equation $x^2 - 3x + 7 = 0$ has roots α and β .
- (i) Find the value of $\alpha + \beta$ 1
- (ii) Find the value of $\alpha\beta$ 1
- (iii) Form the equation which has roots α^2 and β^2 4
- b) A discontinuous function is partly defined as $f(x) = 3x - 1$ for $x > 0$. It is given that $f(x)$ is an odd function.
- (i) Draw a sketch of the graph of $y = f(x)$ 2
- (ii) Complete the definition of $f(x)$ for $x \leq 0$. 2
- c) Show that the locus of points equidistant from the two straight lines $y = 2$ and $3x - 4y = 1$ is a pair of perpendicular lines, and find their equations. 4
- (Hint: the distance of a point from a line is the perpendicular distance)

Question 8 Begin a new sheet of paper

- a) (i) Expand $-(x - 5)(x - 1)$ 1
- For the function $y = \frac{x-3}{(x+1)(x-2)}$ it is given that $\frac{dy}{dx} = \frac{-x^2+6x-5}{(x+1)^2(x-2)^2}$
- (ii) Find any stationary points and establish their nature. 4
- (iii) Find any discontinuities 2
- (iv) Find $\lim_{x \rightarrow \infty} \frac{x-3}{x^2-x-2}$. What is its graphical significance? 2
- b) Find, as a relation between k , l and m , the condition for the quadratic equation in x 4
- $$(k^2 + l^2)x^2 + 2l(k + m)x + (l^2 + m^2) = 0$$
- to have real roots. Give your answer in simplest form.

2005 Solutions to Extension 1 Preliminary

Question 1

$$k:l = 3:-1$$

$$a) \quad x = \frac{3x-1 + -1x-5}{3 + -1} = \frac{-3+5}{2} = 1$$

$$(1, -6\frac{1}{2})$$

$$y = \frac{3x-2 + -1x-7}{3 + -1} = \frac{-6-7}{2} = -6\frac{1}{2}$$

b) multiply b/s by x^2 as it is positive

$$x^3 - 5x^2 \geq 6x$$

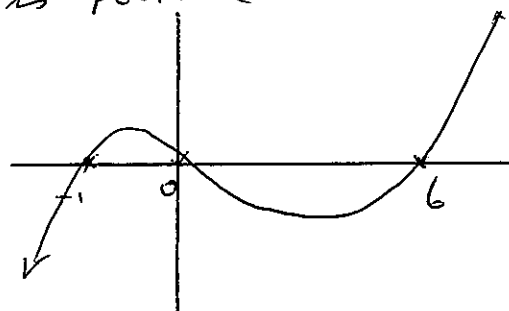
$$x^3 - 5x^2 - 6x \geq 0$$

$$x(x^2 - 5x - 6) \geq 0$$

$$x(x-6)(x+1) \geq 0$$

$$-1 \leq x \leq 0 \quad \text{or} \quad x \geq 6$$

But $x \neq 0 \quad \therefore \underline{-1 \leq x < 0 \quad \text{or} \quad x \geq 6}$



$$c) \quad i) \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta \\ = 2\cos^2 \theta - 1 \\ = 1 - 2\sin^2 \theta$$

$$ii) \quad LHS = \frac{1 - \cos 2\theta}{1 + \cos 2\theta} \\ = \frac{1 - (1 - 2\sin^2 \theta)}{1 + 2\cos^2 \theta - 1} \\ = \frac{2\sin^2 \theta}{2\cos^2 \theta} \\ = \underline{\underline{\tan^2 \theta}} = RHS$$

Question 2

$$a) \quad i) \quad (p+3)^2 - 16p = 0$$

$$p^2 + 6p + 9 - 16p = 0$$

$$p^2 - 10p + 9 = 0$$

$$(p-9)(p-1) = 0$$

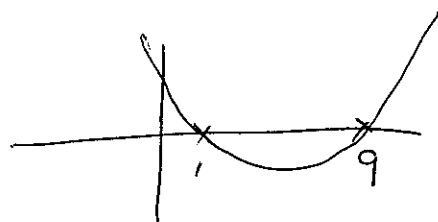
$$\underline{p = 9 \text{ or } 1}$$

($\Delta = 0$ for equal roots)

ii) $\Delta < 0$
(for no real roots)

$$(p-9)(p-1) < 0$$

$$\underline{1 < p < 9}$$



2b) i) $x=2$, $y=6-4=2$ (2,2)

$$\frac{dy}{dx} = 3-2x$$

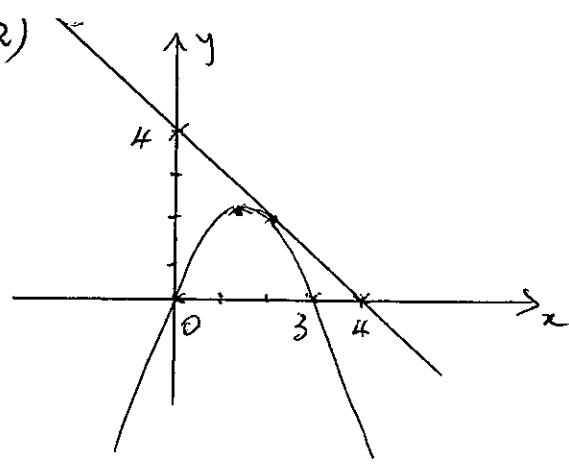
$\therefore$ when $x=2$, $m=-1$

$\therefore y-2 = -1(x-2)$

$$\underline{x+y-4=0}$$

ii) $y = x(3-x)$

vertex $\left(\frac{3}{2}, \frac{9}{4}\right)$ or $\left(1\frac{1}{2}, 2\frac{1}{4}\right)$



c) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$\tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{4+2\sqrt{3}}{2} = \underline{\underline{2+\sqrt{3}}}$$

Question 3

a) Put $u = 3^x$

$$\therefore u^2 + 2u - 15 = 0$$

$$(u+5)(u-3) = 0$$

$$u = -5 \text{ or } 3$$

$$\therefore 3^x = -5 \quad \text{or} \quad 3^x = 3$$

No solution

$x=1$ is only solution.

b) $7x^2 - 5x + 3 \equiv A(x+1)^2 + B(x+1) + C$

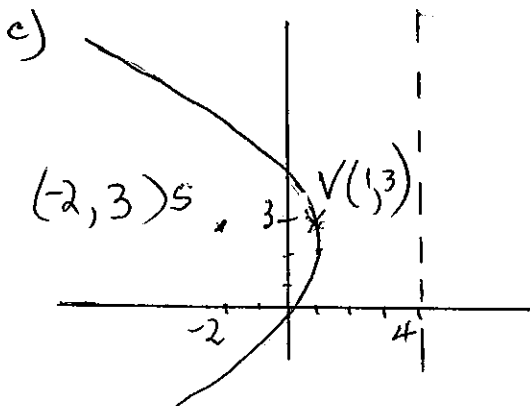
Equate coefficients of x^2 : $7 = A$ ————— ①

Sub $x=-1$: $7 + 5 + 3 = C$

$$C = 15$$
 ————— ②

Sub $x=0$: $3 = 7 + B + 15$

$$B = -19$$
 ————— ③



ii) vertex (1, 3) Form $y^2 = -4ax$

iii) $(y-3)^2 = -4 \times 3 (x-1)$

$$\underline{\underline{(y-3)^2 = -12(x-1)}}$$

Question 4

a) i) $y = \frac{x}{(1-2x)^{1/2}}$

$$\frac{dy}{dx} = \frac{(1-2x)^{1/2} \cdot 1 - x \cdot \frac{1}{2} (1-2x)^{-1/2} \cdot (-2)}{1-2x}$$

$$= \frac{\sqrt{1-2x} + \frac{x}{\sqrt{1-2x}}}{1-2x}$$

$$\frac{dy}{dx} = \frac{1-2x + x}{(1-2x)\sqrt{1-2x}} = \frac{1-x}{(1-2x)\sqrt{1-2x}}$$

ii) $y = 2x^{3/2} - 5x^{1/2} + x^{-1/2}$

$$\frac{dy}{dx} = 2 \cdot \frac{3}{2} x^{1/2} - 5 \cdot \frac{1}{2} x^{-1/2} - \frac{1}{2} x^{-3/2}$$

$$= 3\sqrt{x} - \frac{5}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}}$$

$$= \frac{6x^2 - 5x - 1}{2x\sqrt{x}}$$

OR

$$\frac{dy}{dx} = \frac{(6x+1)(x-1)}{2x\sqrt{x}}$$

b) $f(x+h) = (x+h) - 5(x+h)^2$
 $= x+h - 5x^2 - 10xh - 5h^2$

$$f(x+h) - f(x) = h - 10xh - 5h^2$$

$$\frac{f(x+h) - f(x)}{h} = 1 - 10x - 5h$$

$$f'(x) = \lim_{h \rightarrow 0} 1 - 10x - 5h$$

$$\underline{\underline{f'(x) = 1 - 10x}}$$

c) $y = x^2 - 3x - 7$

$$\frac{dy}{dx} = 2x - 3$$

Put $2x - 3 = 5$

$$2x = 8$$

$$x = 4$$

$$y = 16 - 12 - 7$$

$\therefore$ Point is

$$\underline{\underline{(4, -3)}}$$

Question 5

$$y = x^4 - 4x^3$$

a) i) $\frac{dy}{dx} = 4x^3 - 12x^2$
 $= 4x^2(x-3)$

For Stationary pts $\frac{dy}{dx} = 0$

$$x = 0 \text{ or } 3$$

$$y = 0 \text{ or } -27$$

$$y'' = 12x^2 - 24x$$

When $x=0$, $y''=0$

no further info on (0,0)

$\therefore$ Examine y'

x	0^-	0	0^+
y'	$-$	0	$-$

$\therefore (0,0)$ is a horizontal inflexion.

Nature of (3, -27)

When $x=3$, $y'' = 108 - 72$

positive

$\therefore (3, -27)$ is a

minimum turning point.

ii) Inflection when $y''=0$

$$12x(x-2) = 0$$

$$x = 0 \text{ or } 2$$

already examined.

x	2^-	2	2^+
y''	$-$	0	$+$

Curve changes from concave down to concave up.

$\therefore (2, -16)$ is a point of inflexion

iii) For x intercepts $y=0$

$$x^4 - 4x^3 = 0$$

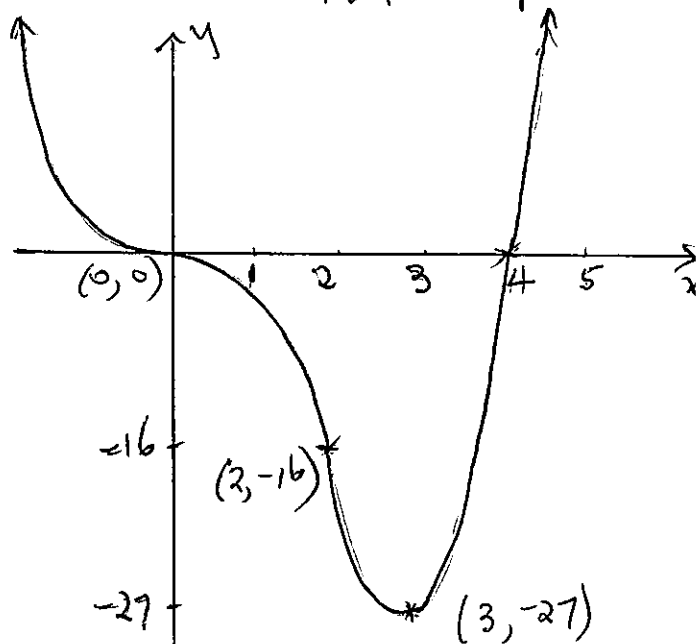
$$x^3(x-4) = 0$$

$$x = 0 \text{ or } 4$$

For y intercepts $x=0$

$$y = 0$$

$\therefore (0,0)$ (4,0) are only intercepts



v) Curve concave down
 $0 < x < 2$

b) locus is

$$(x+5)^2 + (y-4)^2 = (x-3)^2 + (y-2)^2$$

$$x^2 + 10x + 25 + y^2 - 8y + 16 = x^2 - 6x + 9 + y^2 - 4y + 4$$

$$16x - 4y + 28 = 0$$

$$4x - y + 7 = 0 \quad (\text{gradient } 4)$$

Gradient of AB $m_1 = \frac{2-4}{3+5} = -\frac{2}{8}$
 $m_1 = -\frac{1}{4}$

$\therefore$ Perp line has gradient of 4.

Mid point of AB is

$$\left(\frac{-5+3}{2}, \frac{4+2}{2} \right) = (-1, 3)$$

$$4x - y + 7 = -4 - 3 + 7 = 0$$

$\therefore (-1, 3)$ lies on locus.

$\therefore 4x - y + 7 = 0$ is perp bisec

Question 6

a) $\sin(3x - x) = -\frac{\sqrt{3}}{2}$

$0 \leq x \leq 360^\circ$

$\sin 2x = -\frac{\sqrt{3}}{2}$

$0 \leq 2x \leq 720^\circ$

Reference $\angle$ for $2x$ is 60° as $\sin 60^\circ = \frac{\sqrt{3}}{2}$

$\therefore 2x = 180 + 60, 360 - 60, 360 + 240, 360 + 300$
 $= 240, 300, 600, 660$

$x = 120^\circ, 150^\circ, 300^\circ, 330^\circ$

b) i) $2\sqrt{3} \cos x - 2 \sin x \equiv R \cos(x + \alpha)$

$\equiv R \cos x \cos \alpha - R \sin x \sin \alpha$

$\therefore R \cos \alpha = 2\sqrt{3}$ ——— ①

$R \sin \alpha = 2$ ——— ②

② $\div$ ① $\tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$ (as α is acute)

①² + ②² $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 4 \times 3 + 4 = 16$

$R = 4$ as R is positive

$\therefore 2\sqrt{3} \cos x - 2 \sin x \equiv 4 \cos(x + 30^\circ)$

ii) Range of $f(x)$: $-4 \leq y \leq 4$

c) i) $\sin x = \frac{2t}{1+t^2}$ $\cos x = \frac{1-t^2}{1+t^2}$

$\therefore 3 \sin x - 4 \cos x - 4 = \frac{6t}{1+t^2} - \frac{4-4t^2}{1+t^2} - 4$
 $= \frac{6t - 4 + 4t - 4 - 4t^2}{1+t^2}$

$= \frac{6t - 8}{1+t^2}$

ii) First check Does $x = 180^\circ$

$3 \sin 180 - 4 \cos 180 - 4$
 $= 3 \times 0 - 4 \times -1 - 4$
 $= 0$

$\therefore 180^\circ$ is a solution

Then $\frac{6t - 8}{1+t^2} = 0 \Rightarrow 6t - 8 = 0$

$\tan \frac{x}{2} = t = \frac{4}{3}$

$\frac{x}{2} = 53^\circ 8'$ (only $0 \leq \frac{x}{2} \leq 180$ needed)

$\therefore x = 106^\circ 16'$ or 180°

Question 7

a) i) $\alpha + \beta = -\frac{b}{a} = 3$ ii) $2\beta = \frac{c}{a} = 7$

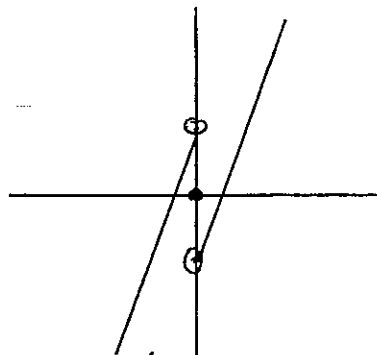
iii) We need $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 9 - 14 = -5$

4 $\alpha^2 \beta^2 = (2\beta)^2 = 49$

$\therefore$ Equation is

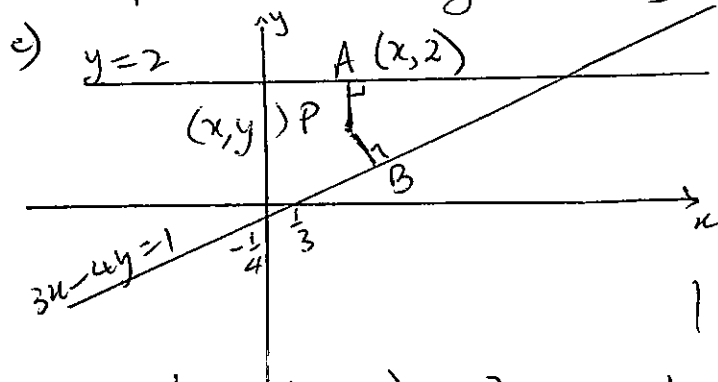
$$x^2 + 5x + 49 = 0$$

b) i)



ii) $f(x) = \begin{cases} 3x-1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ 3x+1 & \text{for } x < 0 \end{cases}$

rotational symmetry about O.



$PA = PB$

$PA = |x-2|$

$PB = \left| \frac{3x-4y-1}{\sqrt{3^2+4^2}} \right|$

$|y-2| = \left| \frac{3x-4y-1}{5} \right|$

$\therefore 5(y-2) = 3x-4y-1$ or $-5(y-2) = 3x-4y-1$

$5y-10 = 3x-4y-1$

$3x-9y-9 = 0$

or $x-3y-1 = 0$

$m_1 = \frac{1}{3}$

$m_1, m_2 = \frac{1}{3}, -3 = -1$

$\therefore$ lines are perpendicular.

$m_2 = -3$

Question 8

a) i) $-(x^2-x-5x+5) = -x^2+6x-5$

ii) $\frac{dy}{dx} = \frac{-(x-5)(x-1)}{(x+1)^2(x-2)^2}$

For stationary points $\frac{dy}{dx} = 0 \therefore \begin{cases} x=5 \\ y=\frac{1}{9} \end{cases}$ or $\begin{cases} x=1 \\ y=1 \end{cases}$

Note denominator of $\frac{dy}{dx}$ is always positive

Nature of $(5, \frac{1}{9})$

x	5^-	5	5^+
y'	$+$	0	$-$

$(5, \frac{1}{9})$ is a maximum

Nature of $(1, 1)$

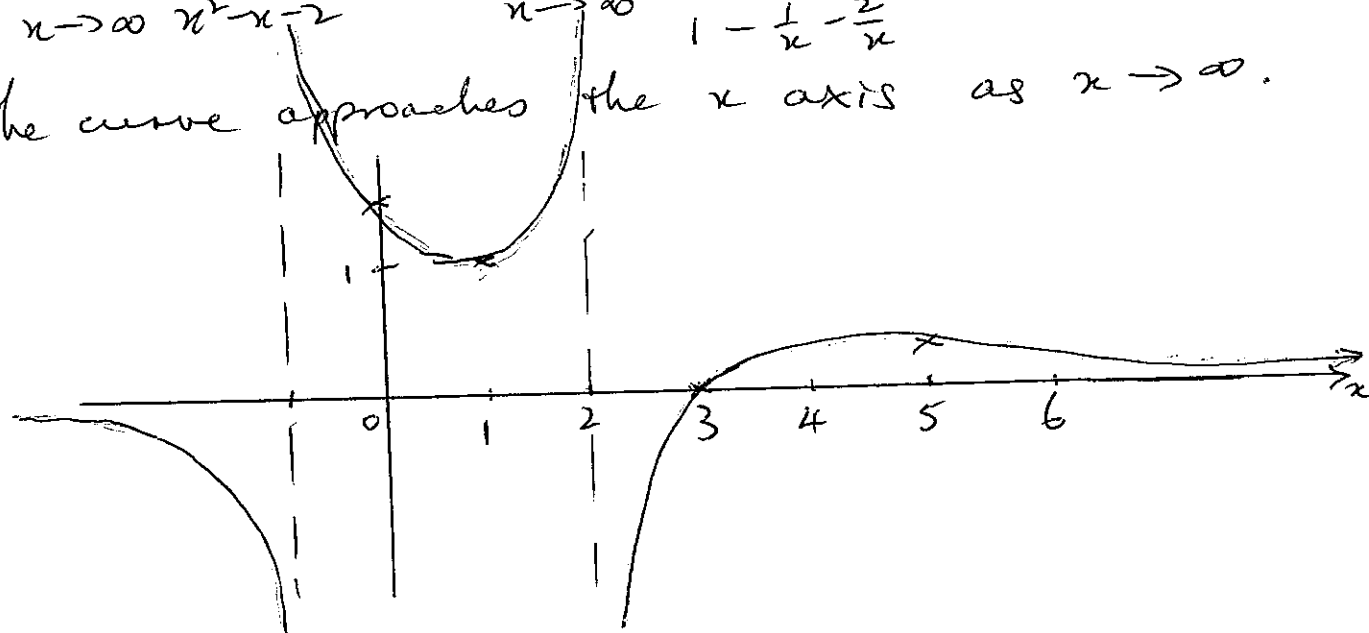
x	1^-	1	1^+
y'	$-$	0	$+$

$(1, 1)$ is a minimum

iii) for Discontinuities $(x+1)(x-2) = 0$
 $x = -1$, $x = 2$ are asymptotes

$$iv) \lim_{x \rightarrow \infty} \frac{x-3}{x^2-x-2} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \frac{3}{x^2}}{1 - \frac{1}{x} - \frac{2}{x}} = 0$$

The curve approaches the x axis as $x \rightarrow \infty$.



b) For real roots, $\Delta \geq 0$

$$4l^2(k+m)^2 - 4(k^2+l^2)(l^2+m^2) \geq 0$$

$$l^2(k^2+2km+m^2) - (k^2l^2+k^2m^2+l^4+l^2m^2) \geq 0$$

$$l^2k^2+2kml^2+m^2l^2 - k^2l^2 - k^2m^2 - l^4 - l^2m^2 \geq 0$$

$$-l^4 + 2kml^2 - k^2m^2 \geq 0$$

$$0 \geq l^4 - 2kml^2 + k^2m^2$$

$$0 \geq (l^2 - km)^2 \quad \text{or} \quad (l^2 - km)^2 \leq 0$$

Which is only possible if $l^2 - km = 0$

$$\text{i.e. } \underline{\underline{l^2 = km}}$$